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Research paper

Multiclass classification of faulty industrial machinery using sound samples

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ABSTRACT

The intelligent diagnosis of faults in industrial assets helps avoid unexpected interruptions of critical services. Thus, machine learning systems for monitoring machinery have a fundamental role in smart factories. The goal of this paper is to promote generalization by combining binary models into multiclass ones. This approach avoids using customized models, which do not perform well if the required conditions do not match during inference. Hence, instead of proposing multiple models for each particular condition, e.g., a model for each machine type, we evaluate different multiclass models where this information becomes an additional outcome along with the operating status. In our experiments, we use the Synthetic Minority Over-sampling Technique (SMOTE) to deal with imbalanced datasets and the PyCaret library to pre-select the most promising algorithms using binary classification. After selecting the binary models for combination, we evaluate two multiclass approaches, called partially- and totally-mixed, using the Malfunctioning Industrial Machine Investigation and Inspection (MIMII) dataset of machinery audio. The number of classifiers needed is reduced up to 83% with our proposal compared with binary classification used as a baseline approach. Furthermore, in the partially-mixed multiclass proposal, it is possible to obtain the correct classification of anomalous states in at least 88% of the samples for most devices. We show that there is a cost on computational resources for model training, which is compensated by the generalization gains verified using the F1-Score metric two times higher than the binary models. We follow the same idea of building more generic models with another existing dataset, ToyADMOS, which similarly contains machine operating sounds. The experiments confirm the results obtained with the MIMII dataset.

1. Introduction

Machine learning algorithms are used in a wide variety of applications (Jiang et al., 2022; Roy and Bhaduri, 2023; Roy et al., 2023; Singh et al., 2023). In the Industry 4.0 context, applications are strongly related to cyber–physical systems, combining edge machine learning solutions and cloud computing to control industrial processes (Silvestri et al., 2020; Aceto et al., 2019). Hence, intelligent solutions emerge to monitor industrial assets, predict and detect faults, and allow for efficient effort management. Given the emphasis on continuous surveillance (Bochie et al., 2021), these possibilities become increasingly important. Consequently, machine learning promotes automated solutions to detect malfunctions and faults within industrial plants, indicating that cyber–physical systems can improve performance by directly impacting industrial productivity (Radziwon et al., 2014; Zhong et al., 2017).

Industrial monitoring often requires interfacing with multiple data sources and types such as vibration signals (Wang et al., 2021; Guo et al., 2018; Lu et al., 2021), images (Li et al., 2018; He et al., 2021), audio samples (Gantert et al., 2021; Natesha and Guddeti, 2021; Sammarco and Detyniecki, 2018), or even a mix of different data (Xie et al., 2021; Huang et al., 2021). This data or part of its features can be used to produce different machine-learning models for reactive maintenance in industrial settings, where each model can be tailored to operate under particular circumstances and scenarios. In related work, for instance, a possible approach is to consider that each new individual machine requires a new model to be trained. This alternative, however, may not scale and may require previous knowledge about the monitored event (Saufi et al., 2019), which is not necessarily available for inference.

This paper proposes the use of multiclass models for reactive maintenance in industrial settings as an alternative to binary approaches. The goal is to promote generalization by combining binary models into multiclass ones. Note that by generalization, we mean reducing the information needed to select the most suitable inference model. Thus, our approach simplifies the use of machine learning for classification

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tasks in production environments. For instance, the type of equipment and the corresponding operational state may be simultaneously identified using a sound sample. This outcome can be obtained without prior knowledge regarding the particular machine in faulty conditions, contrasting with related work.

Thus, by proposing a more generic model, it is possible to implement a fault detection system in more realistic scenarios where multiple machines of different types coexist, and the faulty machine emitting an alarming sound may not be known in advance. It is worth mentioning that the idea does not depend on the dataset but on the existence of samples containing similar data structures.

We evaluate different levels of binary model combinations. We observe that our proposal aligns better with the requirements of smart factories because it does not require a new classifier per individual machine. We propose different multiclass models to investigate the pros and cons of such an approach using the MIMII (Purohit et al., 2019) and the ToyADMOS (Koizumi et al., 2019) datasets. We compare our results considering the binary approach as our baseline (Purohit et al., 2019; Gantert et al., 2021).

For the MIMII dataset, which contains a total of 4 machine types, 4 individual machines, and 3 SNRs, we propose two multiclass models called partially and totally-mixed. Using the proposed models, the number of classifiers considering all types of equipment and SNR levels available is reduced from 48 for the binary classification to 24 in our partially-mixed multiclass proposal and to 8 in our totally-mixed multiclass proposal.

For the ToyADMOS dataset, which contains a total of 3 miniature machine types and 11 individual machines, we follow the same rationale, running experiments using only the binary and the totally-mixed models. The number of classifiers decreases from 11 to 6, confirming the results obtained with the MIMII dataset. Hence, the three main contributions of this paper are summarized as follows:

- we analyze the binary classification with pre-selected supervised-learning models using metrics not used in previous works, such as the Matthews Correlation Coefficient. With the MIMII dataset, we select four algorithms for binary classification using the PyCaret library[1]: Extremely Tree Classifier (ERT), Linear Discriminant Analysis (LDA), Light Gradient Boosting Machine (LGBM), and Gradient Boosting (GB). As the fifth algorithm, we adopt the Support Vector Machine (SVM) because its performance is highlighted in related work (Purohit et al., 2019; Gantert et al., 2021).
- we propose and evaluate multiclass models for reactive maintenance of industrial plants using audio samples. For the MIMII dataset, we propose the partially-mixed model, which requires samples with specific SNR values. The idea is to simultaneously inform the machine type and its current state (whether it is operational or not). The partially-mixed approach also does not require more processing memory than the binary classification. We also propose the totally-mixed model, which mixes samples with different SNRs to approximate a real operating scenario. This approach reduces 83% the number of classifiers and improves the AUC-ROC and F1-Score metrics in the generalization power experiment compared with the binary approach, our baseline model;
- we revisit the same experiments conducted with the MIMII dataset, using the ToyADMOS dataset. The idea is to confirm the conclusions obtained, showing that model generalization is a viable alternative. Nevertheless, instead of using two multiclass model combinations, we only evaluate the more generic one, i.e., the totally-mixed. This experiment validates the possibility of increasing the system generalization without performance losses. The totally-mixed approach reduces the number of classifiers by 45%, maintaining the correct status detection in at least 98% of the cases.

The remainder of this paper is organized as follows. We present in Section 2 the related work. In Section 3, we introduce spectral features extracted and adopted for sound classification. In Section 4, we propose combining multiple binary models into multiclass ones to detect equipment faults, as well as the pre-processing strategy employed. In Section 5, we present the MIMII and the ToyADMOS datasets and the experiments conducted for supervised algorithm selection. Section 6 discusses our experimental results. Finally, we conclude this paper and draw future work in Section 7.

2. Related work

The works selected in this section rely on machine learning algorithms, including neural networks, for event detection. Thus, the existing literature is related to our proposal for at least one of two reasons: (a) being adopted in industrial monitoring or/and (b) using sound samples as input.

Guo et al. (2018) propose a system using a Convolutional Neural Network (CNN) to classify the status condition of bearings from vibration signals. The proposed method comprises a module for status classification and a domain adaptation module to draw inferences on new unlabeled samples by leveraging transfer learning. Vakaruk et al. (2021) uses deep learning to forecast defects in automated guided vehicles mainly used in industry to help industrial processes. The paper proposed an architecture with long short-term memory (LSTM) and CNN in a 5G scenario, adopting the network traffic as input data. Thus, the proposals use neural networks for fault detection, but they neither consider audio samples as input nor the generalization of algorithms for industrial maintenance.

Orrù et al. (2020) propose an approach with MLP and SVM to diagnose faults in the centrifugal pump in the oil and gas industry. The dataset includes data from motor coil temperature, bearings vibration, flow rate, and axial displacement. Chai and Zhao (2019) adopt Random Forest (RF) with feature ranking and selection to analyze static and dynamic features. The method proposed can detect the failure in the Tennessee Eastman process (Bathelt et al., 2015) that contains the reactor, product condenser, vapor–liquid separator, recycling compressor, and product stripper. The algorithm is also successfully applied to detect real-world failures with a flow rate of oil, air, and water in a pressurized system. In this way, both proposals consider supervised algorithms for status classification neither adopting sound samples nor spectral features.

There are proposals in the literature based on machine learning and sound sample classification in different areas. Bui et al. (2020) combine CNN to feature extraction and compare K-Nearest Neighbor (KNN), SVM, RF, and AdaBoost to classify traffic densities in different periods of the day. The visual representation of urban sounds is extracted using Log-Mel Spectrogram and MFCCs. Mohaimenuzzaman et al. (2022) explore the environmental sound classification on microcontrollers, adopting spectrograms and XOR-Networks (Rastegari et al., 2016), a type of CNN with weighs and input assuming binaries values. Then, these proposals adopt sound samples and supervised learning algorithms without focusing on the industrial context or multiclass models.

Pahar et al. (2021) discriminate positive cases of coronavirus disease (COVID-19) using coughs recorded with smartphones as the training set. The MFCCs, MFCCs velocity, MFCCs acceleration, Log frame energies, ZCR, and Kurtosis are the spectral features selected. To address this challenge, the authors compare the performance of SVM and Logistic Regression classifiers in addition to five different neural network architectures. Jamil and Roy (2023) propose the detection of vascular diseases from 1D and 2D cardiac phonocardiogram signal.

[1] Accessed at https://github.com/pycaret/pycaret.

Among the 1D data, the authors select the MFCCs and Linear Prediction Cepstral Coefficients (LPCC) as the input features for the classifiers. The features are selected using bio-inspired methods such as Particle Swarm Optimization (PSO) and Genetic Algorithm (GA). The authors use KNN, SVM, Decision Tree algorithms, and ensemble methods for the classification. Both works consider binary models for detecting diseases using sound samples and extracted features. However, a trained model is required for each disease, which we aim to avoid with multiclass models.

In the literature, instead of computing spectral features, one can convert audio samples into Mel-Spectrogram images to take advantage of all tools available for image processing. Suefusa et al. (2020) conduct experiments using the MIMII dataset with an autoencoder network and a variational autoencoder using Mel Spectrograms as input. Tama et al. (2022) also adopt Mel spectrogram images as input extracted from part of the MIMII dataset. Then, the EfficientNet network is used with a weighted ensemble strategy to detect malfunctions. The authors apply data augmentation techniques to synthetically increase the number of samples and deal with the imbalanced nature of the dataset. The authors in both proposals use Mel Spectrograms as input, as image processing is another alternative for audio processing. These approaches usually adopt convolutional neural networks and, consequently, need to train more parameters than spectral-feature-based proposals, as pointed out by Gantert et al. (2021).

Natesha and Guddeti (2021) compare the performance of MLP, SVM, RF, AdaBoost, and Linear Regression to detect failures in industrial machines, using as input MFCCs or LPC. The authors explore the possibility of implementing the detection system in a fog server as an option for the cloud server to reduce the response time of intelligent Industrial applications. Gantert et al. (2021) propose the adoption of Spectral Features extracted from the MIMII dataset to detect failures. This approach improves failure detection performance compared with the baseline alternative using a Deep Autoencoder. The authors present SVM and MLP as options using only Spectral Features, i.e., without transforming the original samples into Mel Spectrogram images. Gantert et al. (2022) implement the ensemble method SuperLearner to combine supervised algorithms in industrial defects detection. These related works have in common the use of binary classification using the MIMII dataset.

Hence, this paper contributes to the state-of-the-art by proposing a multiclass approach considering spectral features as input, further simplifying industrial plants' training and classification procedures. Unlike related work, our proposal provides more flexibility by combining different binary models with more generic ones. We evaluate different multiclass models and validate our contribution using two datasets, MIMII and ToyADMOS.

3. Spectral features

In this paper, we do not propose new features. Instead, we use spectral features as an alternative to spectrograms, commonly used in the literature (Chaki, 2021). The extracted features are metrics that represent the original audio signals more concisely. We can divide the main features into time-domain, frequency-domain, cepstral-domain, and wavelet based (Sharma et al., 2020). Time-domain features capture signal characteristics over time, while frequency-domain features enable frequency analysis. Frequency-domain features are obtained by converting time-domain features using, for instance, Fourier transform. Cepstral-domain features are related to the cepstrum representation of audio samples obtained through the inverse Fourier transform of the signal spectrum logarithm. Finally, wavelet-based features describe the signal in time and frequency domains and can adopt continuous and discrete characteristics to represent the audio signal.

We reuse the same metrics without loss of generalization to maintain coherency. We efficiently describe sound samples extracting five spectral features previously selected (Gantert et al., 2021). This strategy avoids the use of image recognition algorithms and unsupervised models. The selected features are listed in the following:

- **Mel-frequency cepstral coefficients (MFCC)**: MFCCs describe the instantaneous amplitude of an oscillating signal. Thus, these coefficients are widely used in speech recognition, being computed as the inverse discrete cosine transform (DCT) of cepstrum power at each Mel-frequency. The MFCCs are defined as follows:

$$MFCC(c) = \sum_{m=1}^{M_{mfcc}} X_m \cos\left[c(m-\frac{1}{2})\frac{\pi}{M_{mfcc}}\right], c = 1, \dots, M_{mfcc}, \quad (1)$$

where M_{mfcc} is the number of Mel frequency bands (in our case, $M_{mfcc} = 20$), X_m is the log energy signal in the mth Mel frequency band, and c is the index of the cepstrum coefficient.
The conversion of the frequency into the Mel-frequency scale is:

$$Mel(f) = 2595 \log_{10}\left(1 + \frac{f}{100}\right). \quad (2)$$

- **Spectral Centroid (SC)**: the magnitude of the spectrum center of gravity. This feature is associated with the subjective brightness idea of sound intensity. Consequently, signals with higher frequencies are perceived as clearer. The SC is the average of frequencies present in the signal weighted by their amplitudes:

$$SC = \frac{\sum_{k=1}^{M} f(k)S(k)}{\sum_{k=1}^{M} S(k)}, \quad (3)$$

where $S(k)$ is the spectral magnitude at frequency bin k.
- **Spectral Bandwidth (SB)**: SB is computed considering the spectral centroid SC as follows:

$$SB = \sqrt{\sum_{k=1}^{M} S(k)(f(k) - SC)^2}. \quad (4)$$

- **Spectral roll-off (SR)**: SR is the frequency below which 85% of the spectral magnitude satisfies the following equation:

$$\sum_{k=1}^{SR} S(k) = 0.85 \sum_{k=1}^{M} S(k). \quad (5)$$

- **Zero Crossing Rate (ZCR)**: ZCR counts the number of times an audio signal waveform crosses zero. It is directly computed from the time domain as:

$$ZCR = \frac{1}{2N}\sum_{t=0}^{N-2} |sgn(x(t+1)) - sgn(x(t))|, \quad (6)$$

where $sgn(x(t)) = \begin{cases} 1, & \text{if } x(t) \geq 0 \\ -1, & \text{if } x(t) < 0 \end{cases}$.

Thus, MFCCs are cepstral-domain features; SC, SB, and SR are frequency-domain features; and ZCR is a time-domain feature. We have selected features from the main classes, except wavelet-based, since they combine both frequency and time domains already in use.

4. Fault detection using industrial devices

This paper proposes a fault detection system based on multiclass models for industrial assets using audio samples as input. Using sound-based systems to detect events of interest has many applications. Compared with video-based surveillance systems, a potential candidate is a scenario where the triggering event produces a sound and cannot be seen by a camera. In industrial settings, for instance, malfunctioning equipment may have internal defects hidden from cameras. In this case, using sounds becomes the viable solution. In scenarios like home care assistance, users can feel more comfortable in privacy if only sounds are captured. Babies and elderly people can produce signals in risky situations, such as when they fall or cry (Joshi and Mehetre, 2017). Even in vehicular environments, accidents can be detected by sounds collected by nearby people or other drivers without a camera (Castorena et al., 2023; Sammarco et al., 2024). Hence, the number of

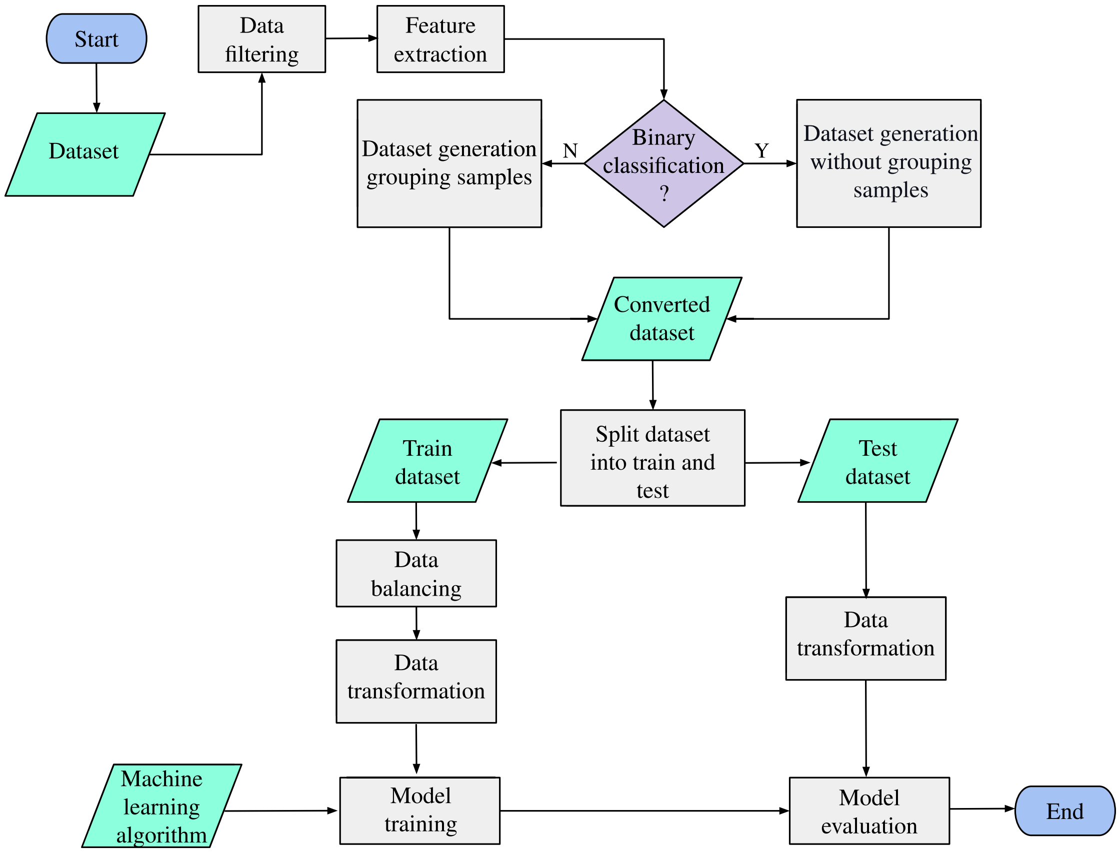


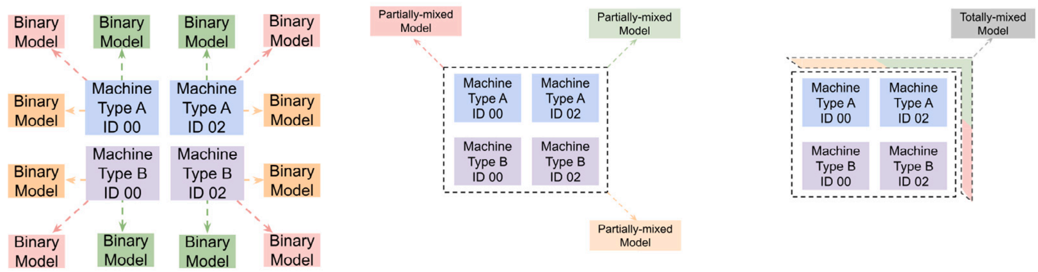
Fig. 1. Workflow for binary and multiclass classifications.

applications that could use sound-based systems for event classification is large, motivating our work. Thus, the proposed system relies on multiclass classification algorithms to deal with the tradeoff between model generalization and the resulting performance. We compare our proposal using two datasets: MIMII and ToyADMOS. We initially use the MIMII dataset to assess the performance of multiclass model employment and then confirm our findings using ToyADMOS. Firstly, we compare the multiclass models derived using the MIMII dataset with related work based on binary classification (Gantert et al., 2022, 2021; Natesha and Guddeti, 2021). Secondly, we reproduce the same idea using the ToyADMOS dataset. This section explains the differences between the binary approach and our multiclass proposals.

The first step in constructing the fault detection system is the selection of machine learning algorithms. To handle that, machine learning models can be empirically evaluated and compared in the same dataset. We consider binary classification to select the most promising algorithms for both classification approaches, i.e., multiclass and binary. Starting with the MIMII dataset, we adopted the PyCaret library as detailed in Section 5. The algorithms with the best performance in binary classification are selected. In addition, we employ the SVM algorithm according to the parameters used by Gantert et al. (2021), considering its performance in the AUC-ROC and F1-Score metrics. For the ToyADMOS dataset, we reuse the same binary models previously selected for comparison purposes.

Upon defining the algorithms, we follow the steps illustrated in Fig. 1. In this figure, green parallelograms, gray boxes, and purple diamonds denote, respectively, inputs and outputs, processes, and decisions. The data filtering stage deals with dataset consistency. For instance, in this stage, outliers can be filtered out, and missing data can be computed using an approximation criterion depending on the workflow implementation. The MIMII dataset is consistent, and we have considered all its samples. In the ToyADMOS dataset, however, we limit the used samples to those with 10 seconds. The MIMII dataset has only 10-second samples, and for coherency, we decided to use samples with the same duration. As seen in Fig. 1, the spectral features are extracted from audio samples in the feature extraction stage. Then, we save the dataset according to the classification approach, as introduced next.

- **Binary classification:** For binary classification, we focus on the design of customized models for very specific conditions. Hence, we use a binary model to detect unexpected behavior, i.e., if the operation is under normal or abnormal status for each machine without grouping samples. For the MIMII dataset, this means considering each machine of a given type under a certain Signal-to-Noise-Ratio (SNR) condition. The MIMII dataset used in this paper organizes the data according to machine type, machine identification, and SNR, allowing model specialization. For the ToyADMOS, this means keeping each machine and its respective case, where the “case” is the machine identifier in the dataset. Note that in the binary classification approach, the idea is to produce multiple customized models for different machines.
- **Multiclass classification:** The multiclass classification focuses on the following issue: having a customized model tailored for a particular setting is not practical in real scenarios. Hence, in our multiclass approach, we reduce the number of models by combining binary models. This reduces the number of classifiers and models if we group samples by leaving aside particular properties. Thus, instead of using different binary models, each one to infer whether a given equipment is operational, we have fewer or only one multiclass model that can provide the same response.
We illustrate the idea in Fig. 2 for the MIMII dataset. Fig. 2(a) shows the binary classification with two different types of equipment (Machine Type A and B) and two machines of each type



(a) Binary classification. (b) Multiclass Partially-Mixed classification. (c) Multiclass Totally-Mixed classification.

Fig. 2. Binary classification and proposed multiclass models focusing on simple design to exchange customization for generalization in industrial sound classifications.

(ID 00 and 02), where the colored arrows indicate different SNR levels. Then, a single individual machine needs a binary model for each SNR. The output of each model is simply an indication of whether the equipment is operating correctly. In the end, the number of binary models is the number of machines types × the number of machines × the number of SNR levels. Fig. 2(b) shows the idea behind the partially-mixed multiclass classification, where devices share the same model but the separation by SNR is still needed. Hence, the used models must be multiclass as they need to output the equipment type and its status. In our example, the number of models is equal to the number of SNR levels, and the number of classes is equal to the number of equipment types times the existing status. Fig. 2(c) shows the totally-mixed multiclass model, where a single model is trained with samples from all the different SNRs. Unlike our previous multiclass proposal, the number of models is unitary as all samples are put together independent of their SNR level.

The output of the multiclass models indicates equipment type and status. However, as seen in the totally-mixed approach, it is possible to reduce the number of models to a single one by removing the need of knowing the SNR level of the received sound sample in advance. This approach increases the generality of the proposal, reducing the number of models to be trained. The motivation comes from the realization that, in an industrial plant, the SNR level of a given audio sample is probably unknown beforehand. In addition, being aware that equipment of a specific type is not correctly working may be enough, contrasting to a more sophisticated model that would identify the particular failing machine. When comparing the three approaches, we started with a total number of models required equal to the number of machine types times the number of machines times the number of SNR levels, which is equal to 48 considering the MIMII dataset. In the partially-mixed approach, the number of models becomes equal to the number of SNR levels in the sound samples, i.e., 3. Then, considering that each model is composed of 8 classifiers, the total number of classifiers adopted is equal to 24. Finally, the totally-mixed approach reduces to a single model composed of 8 classifiers. The motivation for this change is fundamental in an industrial plant, where one may not anticipate the equipment generating the received sound. This knowledge would be required to select the corresponding binary model. In our multiclass approach, we can use a model independent of the generating equipment type. These alternatives provide a more generic property to our proposal, which is desirable in real settings. Note that we mention features like equipment type and SNR level. This is not a requirement but an example with features from the MIMII dataset used in our performance evaluation section (Section 5). The same idea can be reproduced for other datasets, as we show with ToyADMOS.

The totally-mixed multiclass classification further improves the proposal generality by not distinguishing between the different SNR levels. The same idea is used for the ToyADMOS dataset, where we only have the binary and the totally-mixed approach. In this case, we reduce the number of classifiers from 11 in binary classification to 6 in the totally-mixed multiclass approach. We reduce the number of classifiers, aiming to simplify the classification problem. Hence, in this paper, we propose replacing specific binary models with generic multiclass models. The multiclass approach adopts the One-vs-All (also known as One-vs-Rest) strategy, in which the number of classifiers in each model is equal to the number of classes. Each classifier is fitted against all the other classes, similar to a binary classification problem. In other words, the multiclass classification is split into multiple binary problems. The classifiers with higher metric values determine the prediction.

Looking again at the workflow proposed, Fig. 1, the converted datasets are split into training and test sets. The training data has the additional stage of data balancing. This can be implemented by oversampling with the Synthetic Minority Over-sampling Technique (SMOTE) (Chawla et al., 2002) to balance the number of normal and abnormal samples. For this, the algorithm selects a sample from the minority class and their N-nearest neighbors. The neighboring sample is randomly chosen, and a synthetic one is created in the feature space between the selected neighbors. We use N equal to 5 in our experiments. The data balancing stage depends on the dataset, whether it is unbalanced or not, and can be implemented using different techniques. The data transformation step then normalizes the data with Z-Score. Finally, the models are trained with the normalized training data and evaluated with the selected metrics using unseen data from the test set.

5. Experiments

The analysis of the proposed fault detection system uses audio samples from the Malfunctioning Industrial Machine Investigation and Inspection (MIMII) (Purohit et al., 2019) and the Anomaly Detection in Machine Operating Sounds (ADMOS) for miniature machines (ToyADMOS) (Koizumi et al., 2019) datasets. To address this task, we first select the algorithms that achieve higher AUC-ROC (Area Under the Receiver Operating Characteristics Curve) values in binary classification using the MIMII dataset. Then, the best-performing models are applied and compared using the binary and multiclass classification approaches. In this section, we present our experimental setup.

5.1. Datasets

The MIMII dataset contains audio segments of 10 seconds for four equipment types, as shown in Table 1. The ID individually identifies the equipment. Each equipment is of a type, where each type contains 4 individual machines in the MIMII dataset. Thus, it is possible to observe how unbalanced the dataset is. Grouping these samples by equipment type, the number of abnormal samples from machines varies from 2.5%

Table 1
MIMII dataset.

Equipment type	Machine ID	# Normal samples	# Abnormal samples
Fan	00	1011	407
	02	1016	359
	04	1033	348
	06	1015	361
Pump	00	1006	143
	02	1005	111
	04	702	100
	06	1036	102
Slide Rail	00	1068	356
	02	1068	267
	04	534	178
	06	534	89
Valve	00	991	119
	02	708	120
	04	1000	120
	06	992	120

Table 2
ToyADMOS dataset.

Equipment type	Case Number	# Normal samples	# Abnormal samples
ToyCar	1	5400	1056
	2	5400	1060
	3	5400	1060
	4	5400	1060
ToyConveyor	1	7200	1600
	2	7200	1420
	3	7196	1420
ToyTrain	1	5400	1080
	2	5400	1080
	3	5400	1080
	4	5400	1080

to 8.2%, concerning the total number of samples in the dataset. In total numbers, we have 4075 normal and 1475 abnormal samples for fans, 3749 normal and 456 abnormal samples for pumps, 3204 normal and 890 abnormal samples for slide rails, and 3691 normal and 479 abnormal samples for valves. We can observe a variation in the number of samples according to the machine type, in which more information is available for fans. The audio recorded was mixed with factory noises by the dataset's authors, producing three different Signal-to-Noise Ratio (SNR) levels: 6 dB, 0 dB, and −6 dB. Thus, classification with lower SNRs is more challenging.

The ToyADMOS dataset contains audio segments of 10 seconds and 10 minutes for three miniature pieces of equipment: ToyCar, ToyConveyor, and ToyTrain. We conduct experiments adopting only the 10-second samples since this duration is equivalent to the duration used in the MIMII dataset. Table 2 shows the total number of samples used in the ToyADMOS experiments. The number of abnormal samples ranges between 5.4% and 5.8% concerning the total number of samples in the dataset. Also, we have 21600 normal and 4236 abnormal samples for ToyCar, 21596 normal and 4440 abnormal samples for ToyConveyor, and 21600 normal and 4320 abnormal samples for ToyTrain. We then observe that the ToyADMOS dataset provides more samples for each equipment type than the MIMII dataset, and that the number of samples of each type has low variation. Again, there are more normal than abnormal samples. The spectral features are extracted from the audio samples with the Librosa library (McFee et al., 2015). This conversion generates files in CSV format, where each line of the file for an individual machine corresponds to the information of a sound sample from the original dataset. The spectral features correspond to the file columns. We use PyCaret (Ali, 2020) to automate the model selection process. After selecting the binary algorithms, we split the dataset into train and test sets. We randomly select 60% of the data to compose the training set and the remaining samples for the test set. The data transformation and model training are implemented with Scikit-learn (Pedregosa et al., 2011), except the LGBM algorithm, which was implemented with the LightGBM library.[2] The data balancing, in this case the SMOTE, is implemented with the Imbalanced-learn library.[3]

Fig. 3 shows part of the samples in tabular format after the feature extraction stage. The training and testing datasets are composed of samples with a total of 20 MFCCs features plus a label describing the corresponding class.

5.2. Evaluation metrics

We employ the AUC-ROC metric, which provides the relationship between the false positive rate (x-axis) and the true positive rate (y-axis). The value varies between 0 and 1 — the closer to one, the better the performance. The AUC-ROC was selected as the metric to determine the best performing algorithms for binary classification and to evaluate the proposed multiclass models. We use AUC-ROC because it is the metric adopted by the dataset authors who proposed the binary approach we consider the baseline (Purohit et al., 2019). The second metric is the AUC-PR (Area Under the Precision-Recall Curve), where the recall is on the x-axis, and the precision is on the y-axis. Recall (Eq. (7)) and precision (Eq. (8)) are defined as follows:

$$precision = \frac{TP}{TP + FP}, \tag{7}$$

$$recall = \frac{TP}{TP + FN}, \tag{8}$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives. Similarly to AUC-ROC, the AUC-PR metric assumes values between 0 and 1, where 1 denotes a perfect classification. The AUC-PR metric provides a more realistic evaluation for unbalanced binary classifications.

The third metric is the F1-Score, seen in Eq. (9), defined as the harmonic mean between recall and precision. We adopt the macro strategy for multiclass classification, where the F1-Score is computed for each label and the final value is an unweighted mean.

$$F1 = 2 \times \frac{(precision \times recall)}{(precision + recall)}. \tag{9}$$

The last evaluation metric adopted is the Matthews Correlation Coefficient (MCC). MCC can assume values between −1 and 1, where 1 is a perfect prediction, −1 is a completely wrong prediction, and 0 is an almost random prediction. This metric measures the quality of binary classification and is considered more reliable than other metrics, such as F1-Score, accuracy, and Cohen's Kappa (Chicco et al., 2021). The metric suits imbalanced datasets because its formulation considers correct predictions on negative and positive cases, no matter their proportion on the original dataset (Chicco and Jurman, 2020), to achieve high predictive values. MCC is defined as:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}}. \tag{10}$$

5.3. Algorithms selection

The first step in building our multiclass fault detection system is to select the supervised algorithms. Please note that the models are based on supervised learning algorithms, but they are not neural networks, as we have adopted the default algorithms provided by the PyCaret library. The binary classification is implemented following the tutorial

[2] Accessed at https://github.com/microsoft/LightGBM.
[3] Accessed at https://imbalanced-learn.org/stable/references/generated/imblearn.over_sampling.SMOTE.html.

	spectral_centroid	spectral_bandwidth	rolloff	zero_crossing_rate	mfcc1	mfcc2	mfcc3	mfcc4	mfcc5	mfcc6
0	0.214220	0.211259	0.254615	0.207199	-0.086507	-0.286239	0.250552	0.184608	0.664354	1.285931
1	-0.712265	-0.907368	-0.684720	-0.562382	-0.047547	0.804381	-0.653493	0.931011	-0.167001	0.459538
2	-0.821463	-1.128818	-0.876791	-0.538273	-0.099828	1.021329	-0.788328	0.828881	-0.633859	0.201008
3	0.440483	0.777468	0.333885	0.293579	-0.311380	-0.381734	0.482014	-1.930669	-0.573825	-1.317979
4	1.559961	0.961962	1.376494	2.014797	1.269368	-1.893184	-0.943221	-0.270725	-0.520440	0.160918

Fig. 3. Part of the tabular dataset after the feature extraction stage.

obtained from the official PyCaret library repository.[4] We use binary classification as a starting point following the same approach adopted in related work. We employed the SVM algorithm with the hyperparameters tuned by Gantert et al. (2021) and selected the remaining parameters with the support of the PyCaret library.

We select the AUC-ROC as the metric to choose the supervised algorithms, as this is the metric used to evaluate the considered baseline model proposed by the MIMII dataset authors. We select the top-4 algorithms found in this step after sorting them based on the number of times each one has shown the best performance in our experiments (more details in Appendix). We complete the list with SVM, the same model used in related work, for building the multiclass model. The five supervised algorithms selected are then:

- **Extremely Randomized Trees Classifier (ERT):** The ERT is an ensemble method in which randomized decision trees are fitted in the training dataset, and the classification is by majority voting.
- **Linear Discriminant Analysis (LDA):** The LDA algorithm creates a hyperplane to separate the classes adopting the sample features as an axis. The hyperplane is created to maximize the distance between the means of the classes and minimize the variation between the classes.
- **Gradient Boosting (GB):** Gradient Boosting is a boosting ensemble method, i.e., the classifiers are trained sequentially using information from previous classifiers to adapt and improve the final model. In this algorithm, the goal is to minimize the loss function using the gradient parameter to update the weights of each classifier.
- **Light Gradient Boosting Machine (LGBM):** The LGBM differs from Gradient Boosting by using Gradient Based One Side Sampling to reduce the number of samples and the Exclusive Feature Bundling technique to bundle features. Then, the algorithm complexity and the memory consumption are lower, leading the algorithm to become faster.
- **Support Vector Machine (SVM):** Similar to LDA, SVM creates separation hyperplanes to distinguish classes. Nevertheless, the used criterion is to maximize the distance between the hyperplane and the closest sample.

In this way, we can divide the supervised algorithms into two types: hyperplane- and ensemble-based algorithms. LDA and SVM belong to the first type, whereas ERT, GB, and LGBM are ensemble-based. In these three last algorithms, the used decision trees are a combination of classifiers.

6. Results and analysis

In this section, we present and analyze the experimental results obtained for binary classification with the selected supervised algorithms and the implementation of these algorithms for the multiclass proposals. We run our experiments in Google Colab with Intel(R) Xeon(R) CPU @ 2.20 GHz and 12 GB of RAM. Our results are then fourfold:

[4] Accessed at https://github.com/pycaret/pycaret/blob/master/tutorials/Tutorial-BinaryClassification.ipynb.

Table 3
Hyperparameter values for the selected machine learning algorithms.

Algorithm	Hyperparameter	Value
ERT	`criterion` `n_estimators`	Gini 100
LDA	`solver`	svd

LGBM	`n_estimators`
	`learning_rate`
	`boosting_type`
		100
		0.1
		gbdt

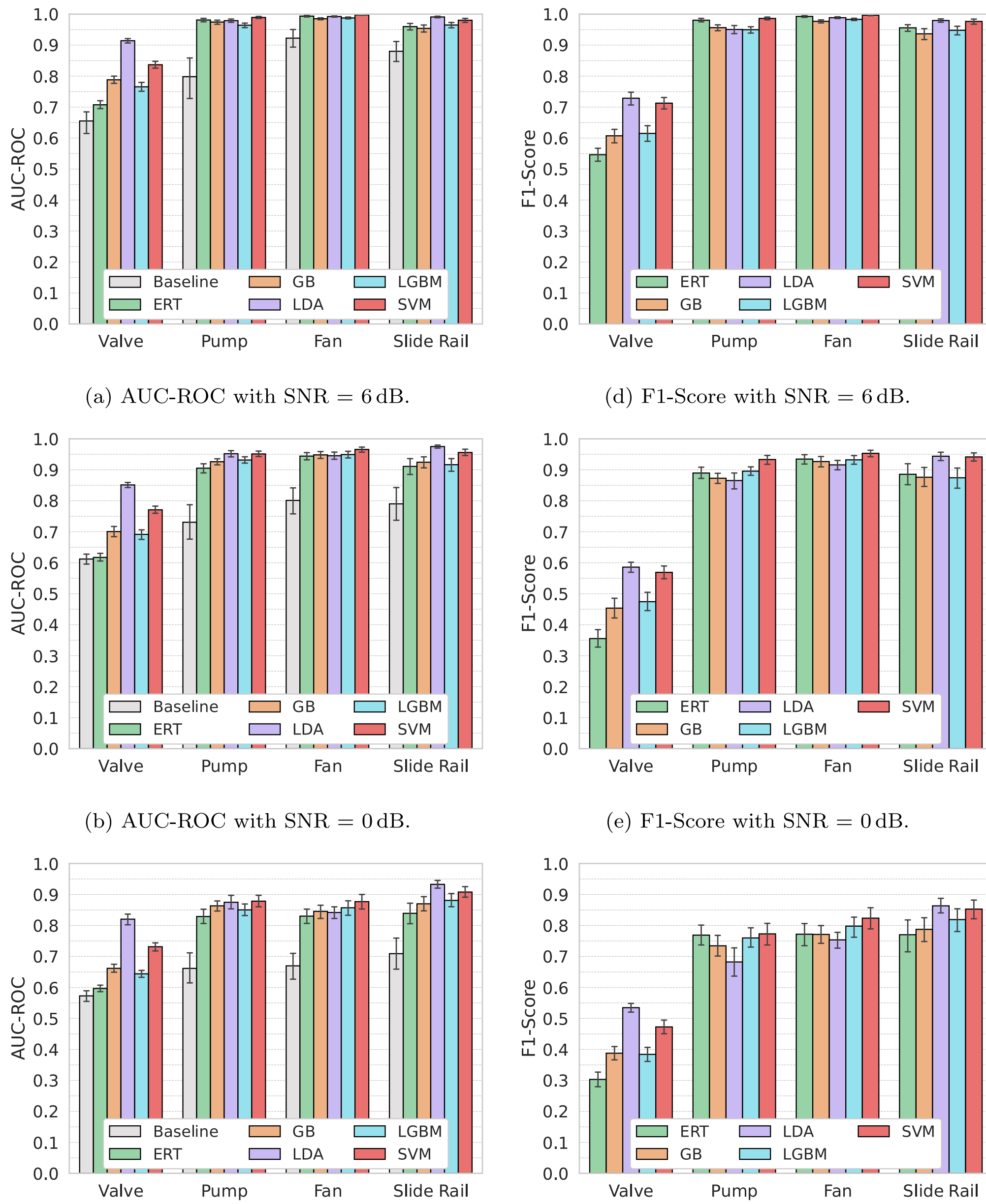
GB	`criterion`
	`n_estimators`
	`learning_rate`
		Friedman_mse
		100
		0.1

SVM	`C` `kernel`	1.0 rbf

(a) we replicate experiments from related work using an automated tool to select the best-performing algorithms. The idea is to observe if performance improvements in binary classification are possible. At the same time, we enlarge the evaluation with more metrics;
(b) we implement the partially-mixed and totally-mixed multiclass proposals to build a more generic and lightweight model to identify the status of industrial equipment with different background noise levels. Then, we compare the proposal with the binary approach considering their classification performance, and finally present the confusion matrices of the proposed multiclass approaches using the algorithms with the best performance in previous results;
(c) we compare the time for model training and the generalization power of partially-mixed and totally-mixed multiclass models compared with the binary model;
(d) in the last experiment, we replicate the experiments in the 10-sec samples from the ToyADMOS dataset to validate the multiclass proposal.

Table 3 presents the most relevant hyperparameters used by the deployed algorithms. These hyperparameters are the default values defined by the PyCaret library, as we use this library for model selection. In ERT, we use the Gini Impurity function; in LDA, we use the Singular Value Decomposition (SVD) solver, which is particularly recommended when the number of features is large. In LGBM, we have gbdt, indicating the use of the Traditional Gradient Boosting Decision Tree as the boosting type; in GB, the criterion to evaluate the quality of a split is Friedman's Mean Squared Error. Finally, in SVM, the kernel rbf denotes the Radial Basis Function.

Before moving to our results, we emphasize the reduction in the number of models. Thus, we compare the number of classifiers needed for the binary classification and for the proposed multiclass approaches using the MIMII dataset. The binary classification requires 48 classifiers considering three SNR levels available, four different equipment types, and four different equipment IDs per type. Each classifier outputs normal or abnormal operational conditions. Regarding the two proposed multiclass models, for the partially-mixed multiclass approach, we have 3 classifiers (one for each SNR level) and 8 distinct classes per multiclass model (four equipment types and two possible outputs — normal or abnormal operation), totaling 24 classifiers. By adopting the totally-mixed, we reduce the number of classifiers to 8 (1/3 compared with the



(a) AUC-ROC with SNR = 6 dB.
(b) AUC-ROC with SNR = 0 dB.
(c) AUC-ROC with SNR = −6 dB.
(d) F1-Score with SNR = 6 dB.
(e) F1-Score with SNR = 0 dB.
(f) F1-Score with SNR = −6 dB.

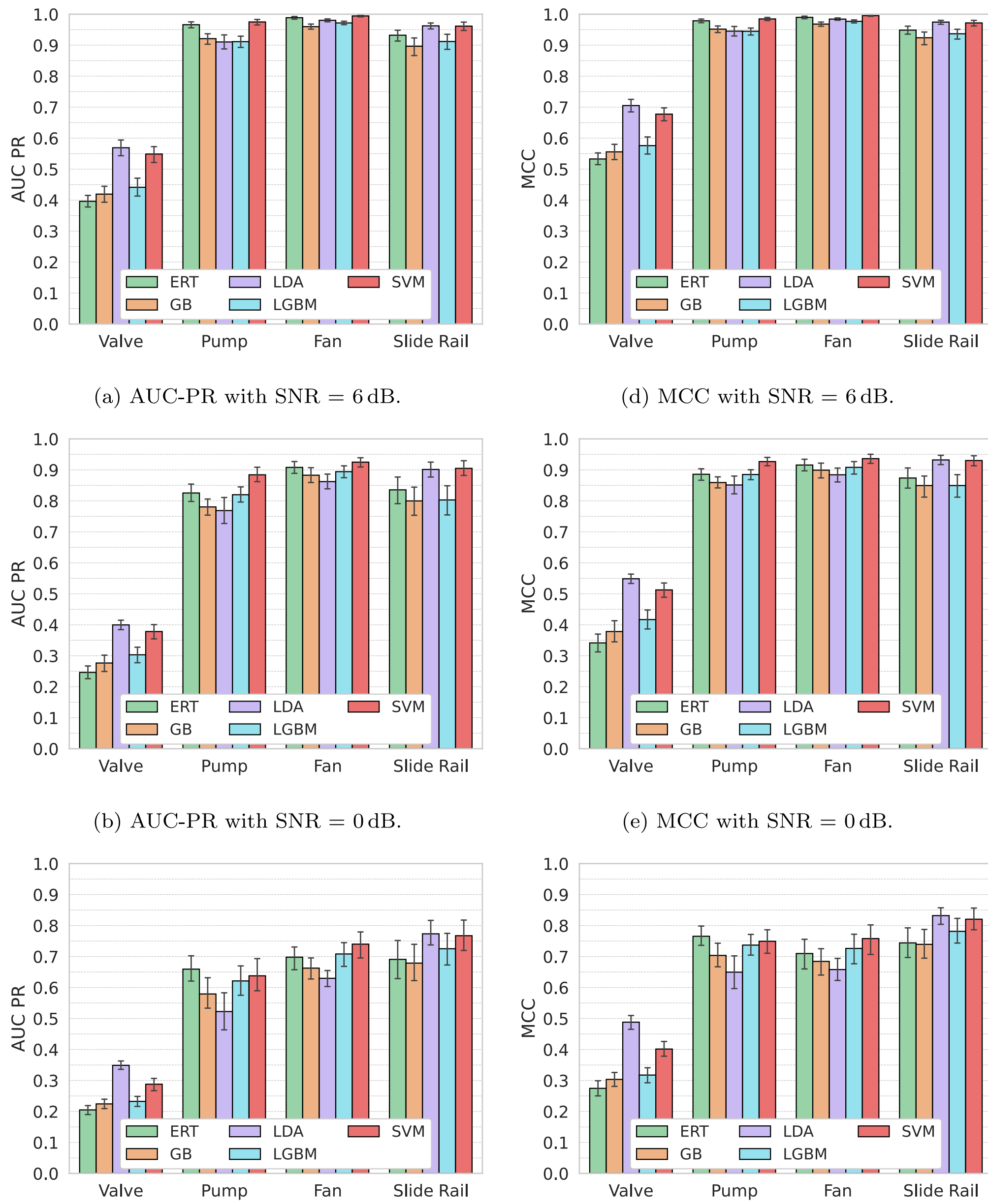
Fig. 4. AUC-ROC and F1-Score for the different algorithms and different SNR values.

partially-mixed). Thus, it is possible to reduce the number of classifiers needed by up to 83%. The dataset already provides the appropriate structure for binary classification, and each model is trained using data from individual machines.

6.1. Binary classification

We start the analysis with AUC-ROC and F1-Score metrics, considering the MIMII dataset and the related work (Gantert et al., 2021, 2022). We divide the results according to equipment type, where the bars are from the different equipment IDs. We consider the results published by the MIMII dataset authors, represented by gray bars, as the baseline. Their model is an autoencoder with five layers: 8 neurons in the bottleneck layer and 64 neurons in the remaining ones. Our results are plotted with a confidence interval of 95%. Figs. 4(a), 4(b), and 4(c) show the AUC-ROC for the three different SNRs, 6 dB, 0 dB, and −6 dB, respectively. Valves, baseline, and ERT (this last one represented by green bars) have similar results, around 0.55 and 0.6 for the different SNRs, indicating low classification quality. Nevertheless, LDA and LR, the purple and the red bars, respectively, achieve AUC-ROC results higher or near 0.8 for all SNRs. These results indicate that hyperplane-based algorithms (LDA and SVM), in addition to improving the performance of the baseline model based on neural networks, achieve higher values for the AUC-ROC metric compared with ensemble-based algorithms. When comparing the performance of the selected algorithms according to the F1-Score metric, we observe that hyperplane-based algorithms stand out for both valves and slide rails in all SNRs, as shown in Figs. 4(d), 4(e), and 4(f). We do not provide results for the baseline approach using F1-Score as the authors only provide AUC-ROC results.

Fig. 5 corroborates the results obtained with AUC-ROC and F1-Score metrics using AUC-PR and MCC metrics. LDA and SVM algorithms



(a) AUC-PR with SNR = 6 dB.

(d) MCC with SNR = 6 dB.

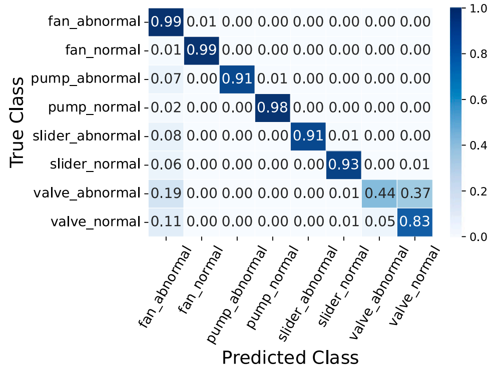
(b) AUC-PR with SNR = 0 dB.

(e) MCC with SNR = 0 dB.

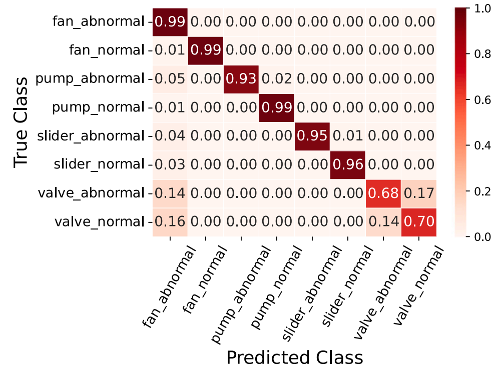
(c) AUC-PR with SNR = −6 dB.

(f) MCC with SNR = −6 dB.

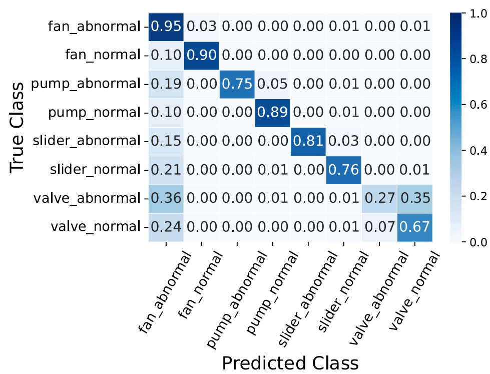
Fig. 5. AUC-PR and MCC for the different algorithms and different SNR values.



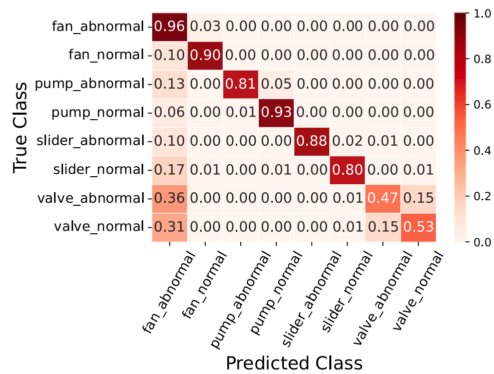
(a) LGBM with SNR = 6 dB.



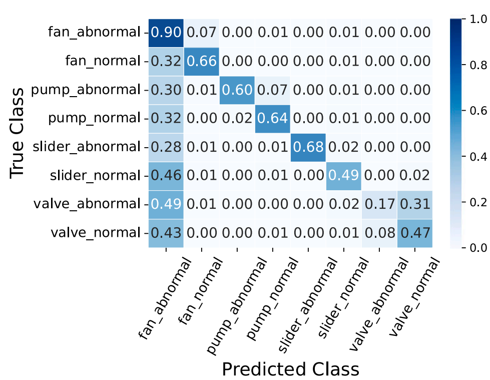
(b) SVM with SNR = 6 dB.



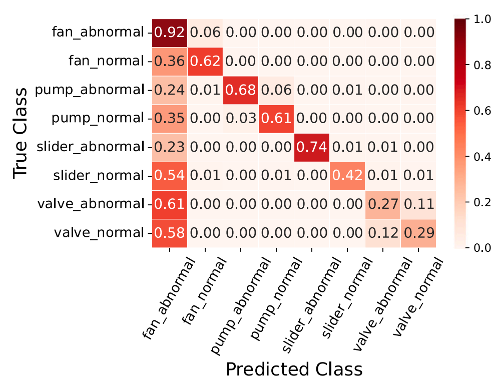
(c) LGBM with SNR = 0 dB.



(d) SVM with SNR = 0 dB.

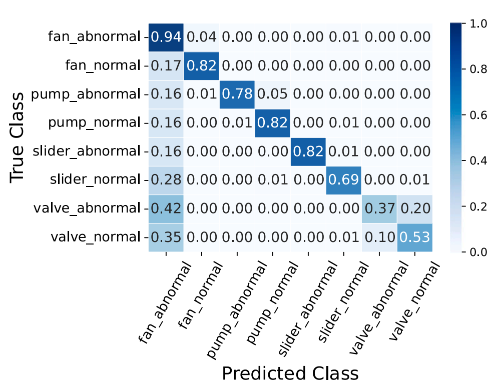


(e) LGBM with SNR = −6 dB.

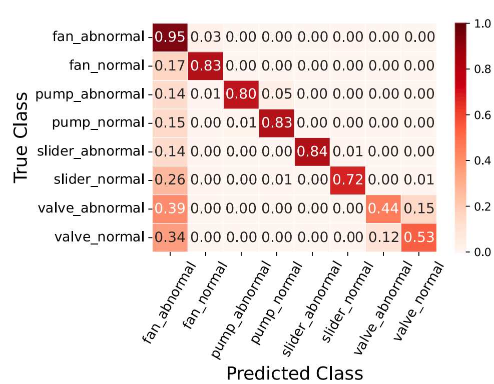


(f) SVM with SNR = −6 dB.

Fig. 6. Confusion matrices for LGBM and SVM for partially-mixed multiclass classification with samples separated by SNR ratios.



(a) LGBM with mixed SNRs.



(b) SVM with mixed SNRs.

Fig. 7. Confusion matrices for LGBM and SVM in totally-mixed multiclass classification using all samples, independent of the SNR ratio.

confirm the superior results for valves and slide rails compared with ensemble-based algorithms for all SNRs. Considering the ensemble-based algorithms, the LGBM represented by blue bars gets higher values for the metrics compared with ERT and GB in the most challenging SNR in Figs. 4(c), 4(f), 5(c), and 5(f). In this way, we select the SVM and LGBM to represent the ensemble-based and hyperplane-based algorithms in multiclass proposals.

6.2. Multiclass classification

We start our evaluation of the multiclass proposals exclusively using the MIMII dataset. The partially-mixed approach keeps the SNR levels separated, while the totally-mixed approach puts together samples with different SNR levels. The totally-mixed multiclass approach aims at reproducing a more realistic scenario, where classification cannot rely on samples separated by SNR levels. We then divide our analysis considering the classification performance, the amount of computing resources for training, and the generalization ability of all models. In this paper, we argue that the multiclass approaches are more general. Hence, the goal is to propose a general-purpose models that can still perform well even in worst-case scenarios. By worst-case scenarios, we mean conducting tests considering the best training conditions for the binary models.

6.2.1. Classification performance

Fig. 6 shows the confusion matrices for LGBM and SVM using the partially-mixed approach. We plot the mean values after 10 rounds. We select these algorithms considering their performance on the overall AUC-ROC and F1-Score metrics. The experiments for SNR equal to 6 dB is presented in Figs. 6(a) and 6(b). LGBM correctly classifies 99% of abnormal class samples for fans, 88% for pumps, and 91% for slide rails. For SVM, the correct classification of abnormal samples achieves results between 94% and 99%. Except for the valve results, the evaluated algorithms can identify the equipment type and the corresponding status for all classes.

For the totally-mixed multiclass proposal, samples with different SNR levels, i.e., 6 dB, 0 dB, and −6 dB, are put together. Even though the idea is to propose a more generic model, this approach has the side effect of increasing the number of samples for the considered classes. In addition, this experiment aims to replicate a more natural scenario where sound samples have different SNR levels. Figs. 7(a) and 7(b) present the confusion matrices for the algorithms LGBM and SVM, respectively. Compared with our previous experiments, the same equipment shows promising results, maintaining a low error rate. The results for fans illustrate this behavior. The model based on LGBM correctly classifies faults in 82% of the cases and, considering SVM, in 71%. We also highlight that LGBM can correctly classify abnormal status for pumps in 82% of the cases. Hence, compared with the partially-mixed multiclass model, the performance dropped approximately 10% in exchange for more generalization and simplification.

6.2.2. Computing resources for model training

In this section, we evaluate a possible tradeoff of the generalization approach. Fig. 12 shows the average training time spent for the models considering all the executed rounds. We compare the partially-mixed and totally-mixed models with the binary approach. The time needed is the sum of all models trained in sequence using the same computing setup. We observe that the training time depends on the algorithm. The time for LDA, LGBM, and ERT is approximately the same for all approaches, but for SVM and GB the time increases faster for the multiclass models. This indicates that, depending on the algorithm, the number of samples has a more evident impact on the algorithm complexity. It is worth mentioning that the training time does not affect the sound-based system as this computation can be run offline (see Fig. 8).

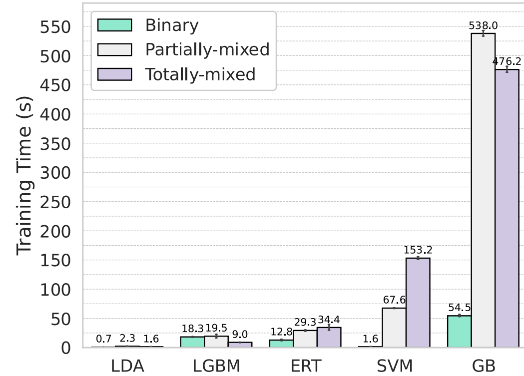


Fig. 8. Training time for the two approaches using the MIMII dataset.

6.2.3. Model generalization

In this experiment, we show the importance of model generalization. The binary models are cross-tested with sound samples from equipment not used for training. For instance, we take the binary model trained for fans and test its performance with a pump sample. Similarly, we cross-test the partially-mixed multiclass models with samples out of the SNR level used for model training. For instance, we test a partially-mixed multiclass model trained for 6 dB samples with −6 dB samples. The single totally-mixed multiclass model does not change, as the training uses the entire dataset.

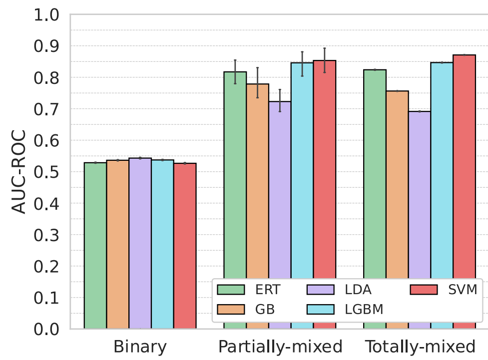
When analyzing the AUC-ROC in Figs. 9(a), 9(b), and 9(c), the binary models do not exceed 0.65 in any of the SNR levels analyzed. On the other hand, the multiclass results reveal that even when tested under more noisy conditions, the LGBM and SVM algorithms can obtain AUC-ROCs of approximately 0.85 at SNRs of 6 dB and 0 dB. Figs. 9(a), 9(b), and 9(c) show the results for the approaches considering the F1-Score metric. The poor generalization of the binary models becomes even more evident due to the AUC-ROC values below 0.3 for all SNR values. As with the previous metrics, the multiclass models achieve superior results for higher SNR levels. However, the totally-mixed multiclass approach outperforms all other alternatives in scenarios with considerable noise, reaching AUC-ROC and F1-Score of approximately 0.8 with the SVM algorithm. This indicates a trend for future research on the power of generic models when particular ones do not suit well, given the amount of noise.

Note that no one expects that a model trained for a certain SNR level would perform well for samples with a different SNR level. Hence, our totally-mixed approach shows that a model trained for all SNR levels performs better than one customized for a particular SNR being used to classify a sound sample arriving under different circumstances. If one cannot guess the SNR level of a received sample in advance, selecting the appropriate customized model is challenging. In this case, our multiclass proposal can help as it brings both the operational status (normal or not) and the equipment type as the outcome independent of the SNR level.

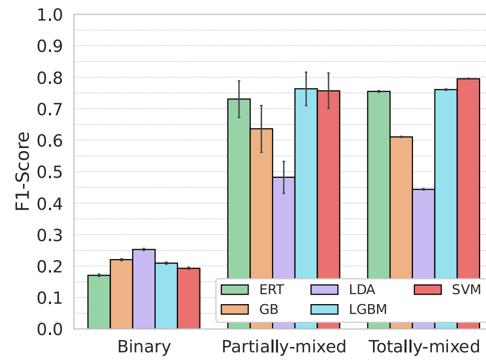
6.3. ToyADMOS experiments

The results presented in this section aim to validate the multiclass proposals previously analyzed with the MIMII dataset. In this way, the steps adopted after selecting the binary models with the PyCaret library are replicated for the ToyADMOS dataset.

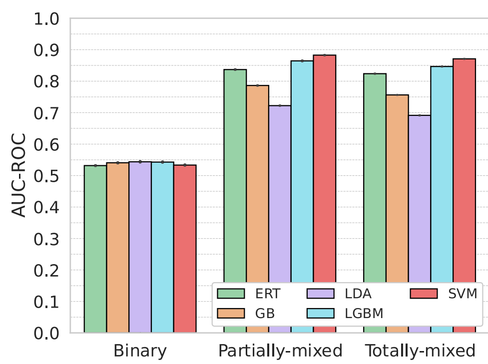
Fig. 10 shows the results obtained for the binary classification. The algorithms selected in the previous steps are also adopted for these experiments. The performance observed for this dataset is superior to that obtained for the MIMII dataset, reaching values above 0.95 for the AUC-ROC metric for all the algorithms and equipment analyzed, as shown in Fig. 10(a). The pattern is repeated for the metrics F1-Score,



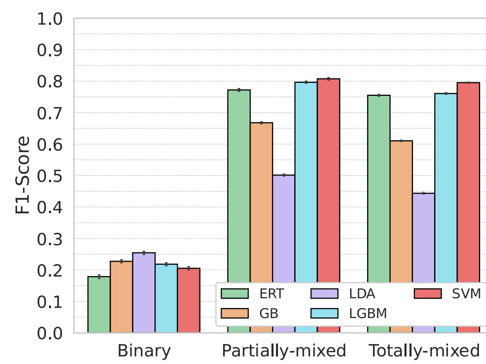
(a) AUC-ROC with SNR = 6 dB.



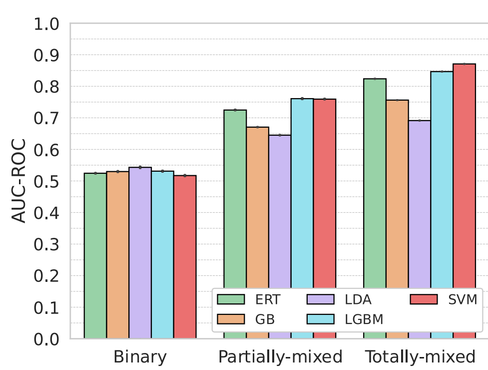
(d) F1-Score with SNR = 6 dB.



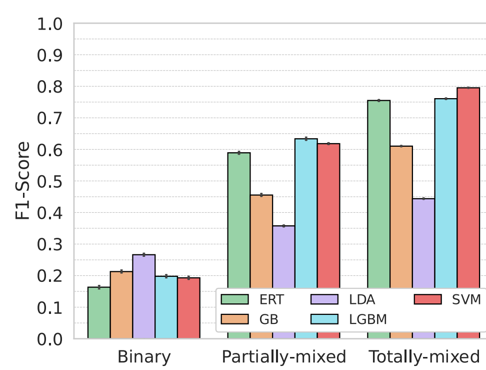
(b) AUC-ROC with SNR = 0 dB.



(e) F1-Score with SNR = 0 dB.

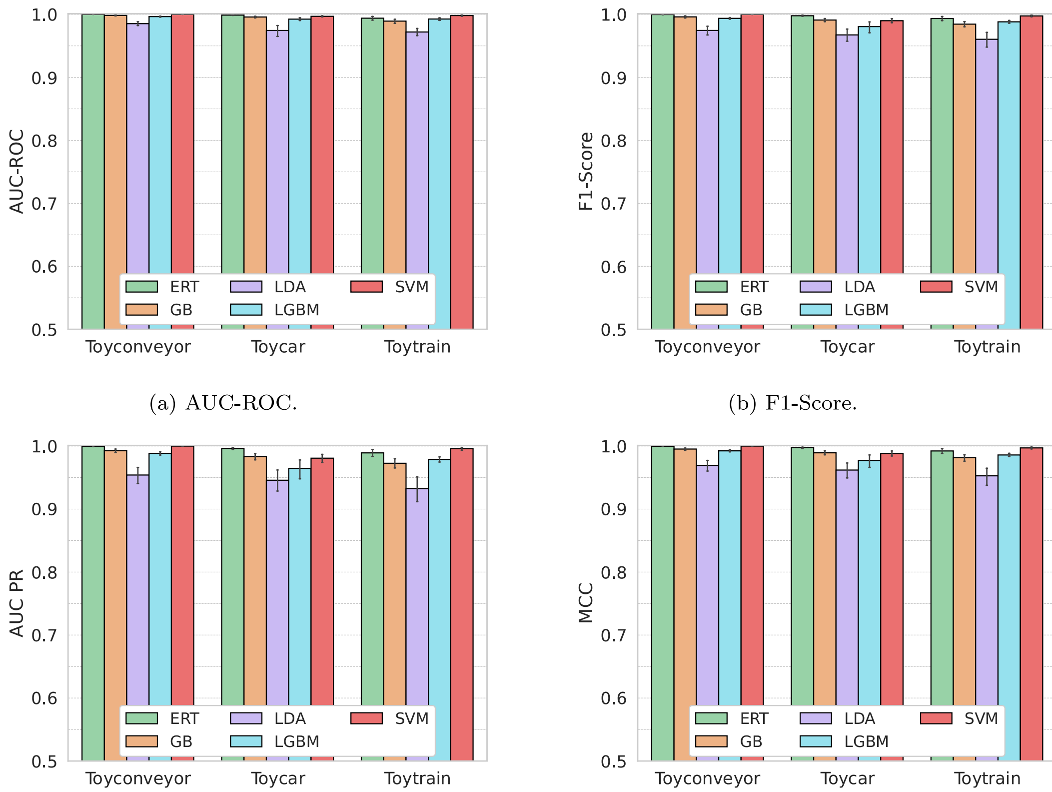


(c) AUC-ROC with SNR = −6 dB.



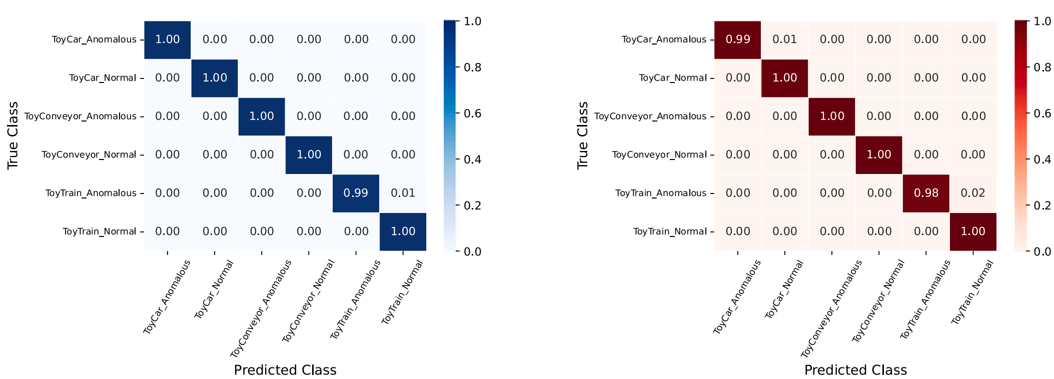
(f) F1-Score with SNR = −6 dB.

Fig. 9. AUC-ROC and F1-Score in the three approaches.



(a) AUC-ROC. (b) F1-Score. (c) AUC PR. (d) MCC.

Fig. 10. AUC-ROC, F1-Score, AUC PR, and MCC for ToyADMOS dataset.



(a) LGBM. (b) SVM.

Fig. 11. Confusion matrices for LGBM and SVM in multiclass classification for ToyADMOS dataset.

AUC-PR, and MCC, in Figs. 10(b), 10(c) and 10(d), respectively. These results show that spectral features and machine learning algorithms are effective for anomalous event detection, even when using other audio sources.

The totally-mixed multiclass approach has also been implemented for the ToyADMOS. Similar to the MIMII dataset, machines of the same type are grouped to create as equipment class with the respective status information. For instance, instead of having a binary model for a particular ToyCar outputting normal or abnormal status, we have a multiclass model outputting the machine type (ToyCar, ToyConveyor, or ToyTrain) and the operational status (normal or abnormal). Fig. 11(a) shows the confusion matrix for the LGBM algorithm. The anomalous status is correctly classified in 100% of the times for the ToyCar and ToyConveyor, and 99% for the ToyTrain. For the SVM algorithm, illustrated in Fig. 11(b), the classification of anomalous status is correct in 100% of the ToyConveyor samples, 99% of the ToyCar samples, and 98% of the ToyTrain samples. The superior performance of the binary models can explain the higher performance of the totally-mixed proposal for the ToyADMOS dataset. Likewise, the main reason for the better performance compared with the MIMII dataset is the greater number of samples available for training.

Fig. 12 shows the time spent on model training. Again, the training time increases in the multiclass proposal, as well as in the previous

Table A.4
AUC-ROC results for binary classification with SNR = 6 dB.

Type	ID	QDA	LDA	NB	LR	ADA	XGB	GB	LGBM	ERT	RF	KNN	DT
Pump	00	**0.774**	0.773	0.768	0.766	0.756	0.755	0.752	0.751	0.749	0.748	0.730	0.726
	02	**0.805**	0.803	0.779	0.798	0.787	0.778	0.780	0.773	0.779	0.780	0.752	0.727
	04	**0.770**	0.761	0.759	0.758	0.736	0.733	0.731	0.729	0.733	0.732	0.735	0.714
	06	0.790	**0.809**	0.806	0.799	0.769	0.778	0.773	0.779	0.771	0.770	0.744	0.737
Slide Rail	00	0.642	**0.654**	0.594	0.651	0.640	0.625	0.646	0.624	0.628	0.625	0.576	0.577
	02	0.658	0.669	0.640	0.663	0.660	0.660	**0.678**	0.662	0.660	0.661	0.613	0.619
	04	0.587	0.629	0.594	**0.630**	0.612	0.624	0.617	0.619	0.611	0.612	0.520	0.555
	06	0.668	**0.688**	0.645	0.683	0.650	0.667	0.659	0.668	0.640	0.646	0.552	0.614
Valve	00	0.686	0.691	0.520	0.641	0.690	0.751	0.747	0.746	**0.755**	0.753	0.673	0.679
	02	0.663	**0.702**	0.569	0.695	0.662	0.682	0.673	0.683	0.680	0.679	0.589	0.613
	04	0.716	0.689	0.571	0.681	0.699	0.728	0.722	0.728	0.730	**0.733**	0.641	0.659
	06	0.696	**0.752**	0.630	0.748	0.733	0.733	0.727	0.737	0.716	0.724	0.608	0.656
Fan	00	0.578	0.591	0.574	0.592	0.592	**0.613**	**0.613**	0.607	0.612	0.609	0.542	0.551
	02	0.633	0.649	0.613	0.642	0.637	0.651	0.655	0.649	**0.656**	0.650	0.576	0.594
	04	0.603	0.642	0.613	0.636	0.646	0.663	**0.673**	0.666	0.670	0.666	0.593	0.605
	06	0.624	0.631	0.605	0.619	0.629	0.653	0.654	0.648	**0.658**	0.654	0.583	0.588

Table A.5
AUC-ROC results for binary classification with SNR = 0 dB.

Type	ID	QDA	LDA	NB	LR	ADA	XGB	GB	LGBM	ERT	RF	KNN	DT
Pump	00	**0.730**	0.716	0.690	0.711	0.717	0.698	0.707	0.697	0.694	0.697	0.667	0.663
	02	0.741	0.731	0.653	0.724	**0.758**	0.743	0.751	0.742	0.742	0.741	0.683	0.684
	04	0.736	0.738	**0.742**	**0.742**	0.697	0.671	0.667	0.671	0.665	0.669	0.677	0.639
	06	0.795	**0.805**	0.778	0.794	0.786	0.780	0.778	0.782	0.778	0.778	0.729	0.752
Slide Rail	00	0.617	**0.629**	0.587	0.626	0.610	0.575	0.610	0.576	0.575	0.574	0.549	0.526
	02	0.646	0.644	0.604	0.637	0.633	0.668	0.660	0.668	**0.671**	0.669	0.594	0.606
	04	0.584	0.621	0.573	0.619	0.602	0.621	0.621	**0.625**	0.624	0.614	0.527	0.564
	06	0.663	**0.705**	0.609	0.689	0.668	0.686	0.675	0.688	0.686	0.686	0.580	0.627
Valve	00	0.599	0.604	0.558	0.599	0.641	0.691	0.677	**0.693**	0.692	0.690	0.608	0.625
	02	0.600	0.637	0.551	0.631	0.627	**0.646**	0.640	0.643	0.639	0.640	0.538	0.577
	04	0.629	0.607	0.547	0.573	0.634	0.715	0.697	0.708	**0.717**	0.715	0.617	0.634
	06	0.669	0.698	0.573	0.694	0.694	**0.722**	0.712	0.717	0.721	0.719	0.621	0.647
Fan	00	0.559	0.574	0.553	0.574	0.569	0.592	0.584	**0.594**	0.591	0.589	0.532	0.528
	02	0.584	0.599	0.571	0.597	0.608	0.610	**0.620**	0.613	0.612	0.608	0.551	0.558
	04	0.589	0.625	0.597	0.624	0.617	0.619	**0.629**	0.622	0.617	0.617	0.552	0.561
	06	0.583	0.601	0.580	0.598	0.594	0.615	**0.617**	0.614	0.616	0.609	0.557	0.556

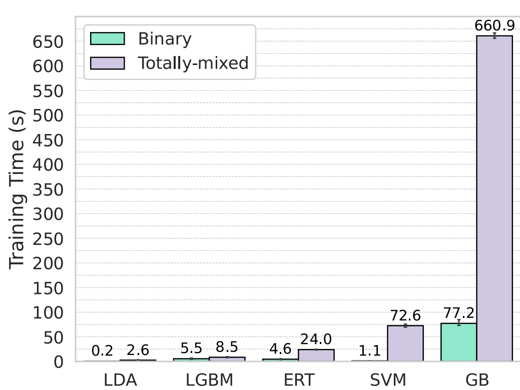


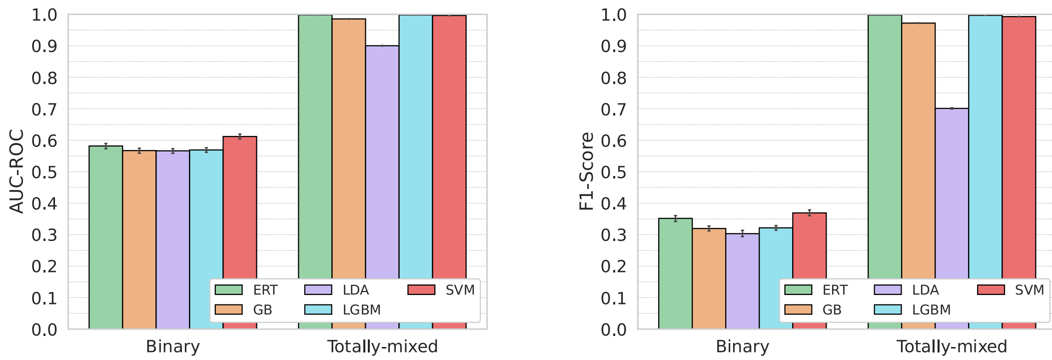
Fig. 12. Training time for the two approaches using the ToyADMOS dataset.

experiment. The cross-test evaluation to measure the model generalization is shown in Figs. 13(a) and 13(b). We compare the binary and multiclass classification using the AUC-ROC and F1-score metrics, respectively. As observed in the previous experiments, the binary classifiers achieve relatively lower performance metrics than the multiclass model when tested with samples from distinct machines. It highlights how specific binary models are. For AUC-ROC the value is just over 0.6 and remains below 0.4 for F1-Score. Meanwhile, the totally-mixed model achieves the maximum value in ERT, LGBM, and SVM algorithms, since these models are trained with all the available data.

7. Conclusion

In this paper, we proposed multiclass approaches to detect abnormal samples in industrial machines using sound samples. The idea is to reduce the number of classifiers and increase the model generality by combining binary models typically used in the literature. The binary classification was a starting point for selecting two algorithms to be used in our multiclass proposals: SVM and LGBM. We evaluated the proposed multiclass models comparing them with the binary alternative using two industrial sound datasets: MIMII and ToyADMOS. We proposed different levels of binary model combinations. One, called partially mixed, groups part of the collected samples using features containing different categories to train less multiclass models. The other combination level proposed, called totally-mixed, groups all data to train a single multiclass model composed of a number of classifiers equal to the number of all considered classes. Our results show that our proposals decrease the number of classifiers between 45% and 83% considering both datasets.

Our proposals have also improved the fault-detection system generality. Our experiments with the MIMII dataset showed that the totally-mixed multiclass model reaches up to 93% of successful classifications. We also observe an improvement up to 58.18% of AUC-ROC and 220% of F1-Score for the totally-mixed multiclass model compared with the binary solution in the model generalization experiment. For the ToyADMOS dataset, we have only evaluated the totally-mixed approach. In this case, the anomalous samples were classified correctly in at least 98%, considering the equipment and algorithm with the lowest success rate. The training time has shown to become a tradeoff that is worth investigating.



(a) AUC-ROC. (b) F1-Score.

Fig. 13. AUC-ROC and F1-Score in the two approaches for ToyADMOS dataset.

Table A.6
AUC-ROC results for binary classification with SNR = −6 dB.

Type	ID	QDA	LDA	NB	LR	ADA	XGB	GB	LGBM	ERT	RF	KNN	DT
Pump	00	0.692	0.707	0.664	0.702	0.712	0.714	**0.716**	0.710	0.696	0.710	0.676	0.663
	02	0.708	0.699	0.600	0.695	0.751	0.774	0.776	0.774	**0.779**	0.775	0.672	0.717
	04	0.738	**0.755**	0.737	0.750	0.712	0.708	0.709	0.709	0.699	0.704	0.639	0.640
	06	0.741	**0.778**	0.716	0.775	0.759	0.774	0.774	0.776	0.769	0.771	0.690	0.724
Slide Rail	00	0.612	0.622	0.598	0.620	0.624	0.651	0.652	0.646	**0.653**	0.649	0.589	0.586
	02	0.620	0.629	0.592	0.626	0.642	**0.682**	0.669	0.681	0.678	0.678	0.597	0.612
	04	0.557	**0.614**	0.559	0.610	0.574	0.571	0.566	0.566	0.551	0.554	0.485	0.493
	06	0.622	**0.665**	0.520	0.652	0.614	0.651	0.629	0.641	0.651	0.642	0.539	0.585
Valve	00	0.592	0.600	0.526	0.586	0.628	0.704	0.696	**0.705**	0.698	0.702	0.626	0.637
	02	0.607	0.653	0.545	0.651	0.634	**0.686**	0.658	0.677	0.679	0.676	0.543	0.613
	04	0.624	0.631	0.535	0.607	0.657	0.708	0.696	**0.710**	0.709	0.706	0.640	0.646
	06	0.691	0.695	0.566	0.692	0.721	0.735	0.728	**0.738**	0.735	0.736	0.640	0.644
Fan	00	0.530	0.538	0.519	0.536	0.526	0.549	0.543	0.550	**0.556**	0.553	0.495	0.499
	02	0.572	0.581	0.541	0.577	0.588	0.604	0.601	0.606	**0.607**	0.606	0.539	0.554
	04	0.584	0.607	0.54	0.606	0.596	0.632	0.622	0.632	**0.636**	0.631	0.546	0.568
	06	0.568	0.582	0.562	0.581	0.585	0.601	0.601	**0.605**	0.601	0.603	0.526	0.542

Our proposal has two main limitations. First, it relies on labeled data for supervised machine-learning training. Labeled data requires annotation procedures, which can be overwhelming if done manually. Even though challenging, this is a relatively known issue addressed in the literature using different approaches, such as semi-supervised learning or pseudo-labeling. In addition, the models proposed provide information for reactive maintenance but do not address failure prediction. Quickly identifying malfunctioning equipment allows resource savings due to more efficient maintenance procedures. Predictive maintenance, however, would further enhance the maintenance process in real factories. The problem for predictive and reactive maintenance is the lack of available datasets. Hence, an additional limitation of our work concerns dataset availability.

In future work, we plan to implement our proposal in production scenarios and improve the multiclass approach. The idea is to construct a more generic and lightweight system to enable the employment of the proposed models in real smart factories with IoT devices. We also plan to carry out a direct comparison with spectrogram-based methods with neural networks usually adopted for image classification tasks. Finally, we plan to perform an ablation study to expand our model evaluation.

CRediT authorship contribution statement

Luana Gantert: Writing – original draft, Visualization, Software, Investigation, Data curation, Conceptualization. **Trevor Zeffiro:** Writing – review & editing, Validation, Formal analysis. **Matteo Sammarco:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Miguel Elias M. Campista:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Luana Gantert reports financial support was provided by Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data is already public.

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Appendix. PyCaret results

Tables A.4, A.5, and A.6 show AUC-ROC results for the binary classification with MIMII dataset by PyCaret library. We consider the different SNR levels and the AUC-ROC as the metric to select the supervised algorithms, as this is the metric used to evaluate the considered baseline model proposed by the dataset authors. We used the top-4 algorithms from this step, sorted by the number of times each one has shown the best performance in our experiments, to complete the five most promising ones for building the multiclass model and the additional experiments using the ToyADMOS dataset.

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